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ECOPULSE

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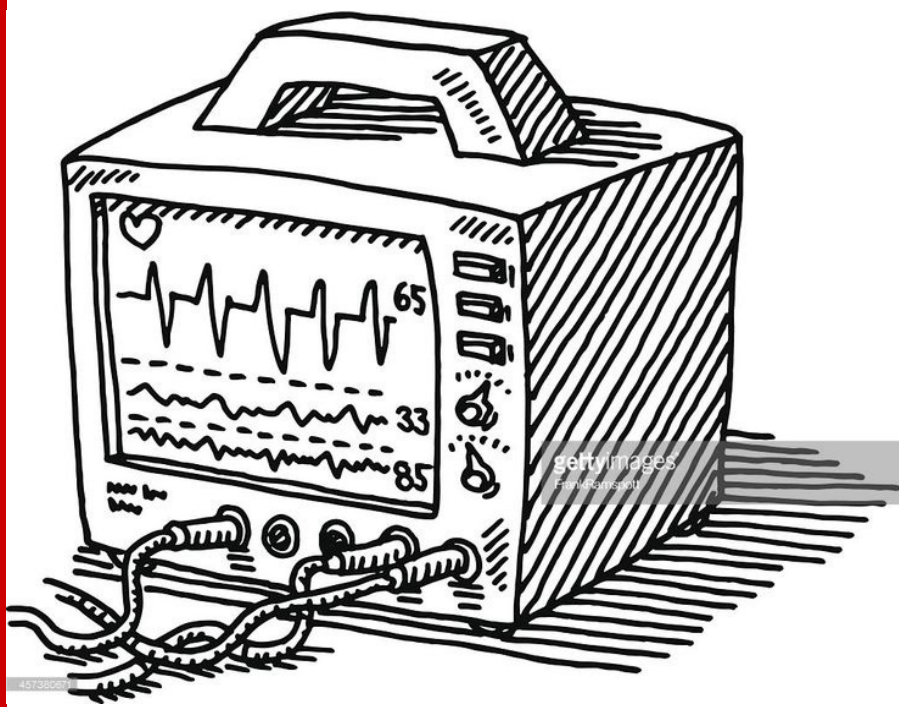


Table of Contents

Abstract 4

Introduction 5

Materials and Methods 8

Table 1: materials used in the implementation of the prototype: 8

Methods 9

Test plan 11

Results 12

Negative Results: 12

Positive Results: 12

Table 2: BPM and SPO_2 readings: 13

Analysis 14

Table 3: Learning outcomes utilized during the project: 14

Photoplethysmography 15

Amplifying and noise reduction 18

Phototransistor vs. Photodiode 19

Accuracy Validation & Mechanical Optimization 20

Conclusions 22

Recommendation 24

Empowering Education and Healthcare through Recycled Innovation 24

Integration of IoT and Telemedicine Capabilities 25

Cuffless Blood Pressure Estimation (PTT) 26

Abstract

Egypt has faced many grand challenges in recent decades, particularly in the quality of education and environmental management. This project aims to solve some of these challenges, including improving the scientific environment for all and recycling waste for economic purposes. These challenges were worsened by the high cost of laboratory equipment, which leaves many public schools relying on passive teaching methods, and the growing hazard of electronic waste (e-waste). The purpose of this study is to address the lack of practical biological monitoring tools in classrooms by transforming e-waste into valuable scientific instruments. The selected solution is a low-cost, eco-friendly heart rate and oxygen sensor based on the principle of Photoplethysmography (PPG). This device utilizes recycled components, including infrared LEDs extracted from an old mouse and photodiodes from discarded scanners, to detect blood volume changes and calculate heart rate. The prototype underwent extensive testing to ensure it meets accuracy standards (within $\pm 5\%$) and portability constraints, as it showed an accuracy of 99% measuring the SPO_2 and 97.9% measuring the BPM compared to the commercial device, with only a margin of error of 1% for the SPO_2 and 2.1% for BPM, offering a reliable alternative to expensive commercial sensors. This project contributes to solving Egypt's challenges by bridging the gap between theoretical and practical

STEM education while promoting sustainability through the innovative recycling of electronic waste.

Introduction

Egypt strives to achieve sustainable development, yet it faces significant challenges that impact its scientific progress and environmental health. Furthermore, among these is the Egypt Grand Challenge to "Improve the scientific and technological environment for all". This challenge is compounded by the "Recycle garbage and waste" challenge, specifically concerning the massive accumulation of electronic waste. In 2022 alone, Egypt generated an estimated 690 kilotons of e-waste, as shown in Figure 1, one of the highest rates in Africa. While this waste poses a severe environmental threat due to toxins like lead and mercury leaking into the soil, it also represents a hidden treasure of valuable electronic components currently being lost to informal dumping.

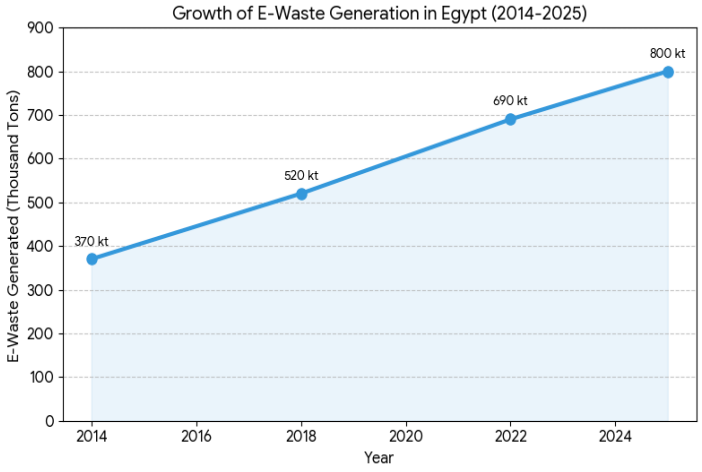


Figure 1: Growth of E-Waste Generation in Egypt (2014-2025)

The specific problem to be solved is the lack of accessible biomedical monitoring equipment in Egyptian public schools and rural healthcare sectors due to

financial constraints. With reports from UNICEF and GAEB indicating that 20% of school buildings are unfit for use, and even advanced institutions like the STEM High School for Boys in 6th of October relying solely on diagrams and microscopes, schools often lack essential physiological sensors for biology education because commercial units cost between 500 to 2,500 EGP, a price that becomes prohibitive when equipping entire classrooms. This lack of equipment forces a reliance on rote memorization rather than active, practical learning. Simultaneously, the mismanagement of e-waste creates a toxic hazard, whereas these discarded devices could provide the components needed to bridge this equipment gap.

To understand how to address this, prior solutions were analyzed. One notable prior solution is the Smart Heart Rate Monitoring System Using Arduino-Uno (2023). A major strength of this system is its portability and low cost, as it uses battery power and simple circuitry, making it deployable in non-clinical settings. However, a significant weakness of this prior solution is its reliance on commercially available modules (such as the MAX30100 sensor) rather than recycled components, which undermines its sustainability and alignment with e-waste reduction goals.

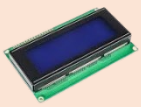
Transitioning to the solution, our project aims to design a Heart Rate and Oxygen Sensor by extracting and reusing functional components from old electronic devices. This prototype will utilize the principle of photoplethysmography (PPG) to measure blood volume changes. By repurposing optical sensors and

microcontrollers found in discarded electronics, such as old printers or computer hardware.

To evaluate the success of the prototype, specific design requirements have been established based on the Capstone Challenge. The prototype must achieve an accuracy within ±5% compared to a standard device. It must utilize at least one extracted electrical component from e-waste, and the housing must be made of transparent material to visualize the internal design. Furthermore, the device must be portable, safe to operate, and capable of displaying real-time physiological data to support active learning in biology laboratories.

Materials and Methods

Table 1: materials used in the implementation of the prototype:

NAME	DESCRIPTION	QUANTITY	IMAGE
ATMEGA328P	The microcontroller is used to calculate the heart rate and oxygen saturation.	1 module	
1KL3A PHOTOTRANSISTOR	Detects changes in light intensity transmitted, allowing the device to measure pulse and SpO_2.	1 module	
RED LED	Emits light at a wavelength which gets absorbed differently by oxygenated and deoxygenated blood.	1 led	
IR LED	Emits infrared light that penetrates the finger, allowing the phototransistor to detect blood volume changes to calculate heart rate.	1 led	
LCD DISPLAY	Used to show the real-time pulse rate and SpO_2 values to the user.	1 lcd	

BREADBOARD	To connect all the components of the heart rate and oxygen saturation monitoring system without soldering.	1 unit	
RESISTORS	Used to limit current and protect components.	8 units	
FINGER CLIP	Holds the LEDs and phototransistor in the correct position to ensure stable and accurate measurements.	1 clip	

Methods

1. System Design and Planning:

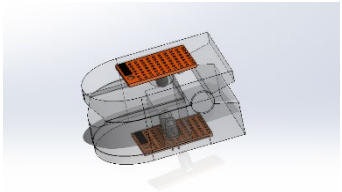


Figure 2: 3D design of finger clip and sensor placement

The project started with meticulous planning and conceptualization to outline the prototype's architecture. Where a 3D design was made to improve the workflow.

2. Component Extraction and Sensor Preparation:



Figure 3: Red LED found in a

To begin with, an old infrared remote control was disassembled as shown in **figure 4**, from which a working IR LED was extracted and cleaned for reuse. Then an old mouse was opened to get the red led inside it as shown in **figure 3**. A damaged receiver

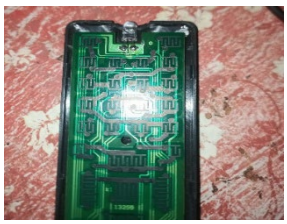


Figure 4: IR from old TV

module was also opened to obtain a functional phototransistor. Each extracted component was tested individually, from which the LEDs and phototransistor



Figure 5: Fake ps5 board

were arranged in a way to ensure stability and high efficiency of heart-rate and oxygen saturation monitoring. Additionally, an ATmega328p from an old fake PS5 board was salvaged, carefully resoldered, and mounted for integration in the next step as shown in **Figure 5**.

3. Sensor Housing:

A small enclosure was prepared from a clip, then got sprayed black to ensure that the light error was minimal. The extracted components were fixed inside the housing, and the wires were soldered and routed through the openings, as shown in **Figure 6**.



Figure 6: Finger clip

4. Schematic Circuit's Design:

Kicad was used to design a circuit containing the ATmega328p microcontroller, current-limiting resistors for the IR and red LEDs, and the phototransistor connected in a common-emitter configuration, as demonstrated in Figure 7. The

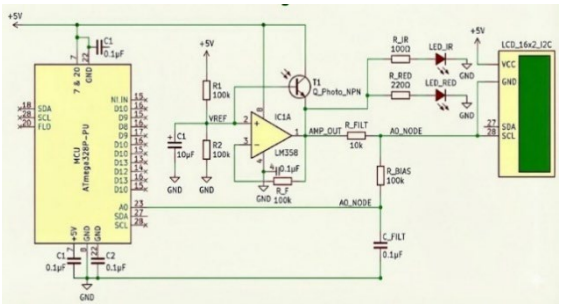


Figure 7: Schematic Circuit's Design

LEDs, phototransistor, and power supply connections were then fixed onto a breadboard and connected to the microcontroller.

Test plan

The accuracy of the device was tested by comparing the Heart Rate (BPM) and SpO2 readings of the prototype against a standard commercial pulse oximeter. Readings were taken from five different individuals to verify consistency. The percentage error was calculated for each trial to ensure that the accuracy deviation did not exceed the required limit of ± 5%.

The sensing mechanism was tested by placing a finger between the extracted IR LED and the photodiode while powered by the battery. The signal was monitored on the LCD screen to ensure the e-waste components could successfully detect blood flow changes and generate a stable Photoplethysmogram (PPG) waveform without significant noise interference.

The physical constraints were tested by placing the entire circuit and housing inside the transparent container. This was done to ensure that the internal recycled components were clearly visible for evaluation.

Results

Negative Results:

Initially, a photodiode was used instead of a phototransistor in the early prototype phase. However, due to its lower internal gain, it produced weaker signals that were highly affected by irregular light interference, reducing the accuracy of the heart rate and oxygen saturation measurements. It also suffered from motion artifacts caused by finger movement, which distorted the PPG waveform and prevented stable readings. Furthermore, the IR and red LEDs were not properly current limited, resulting in insufficient light penetration through the finger. Additionally, the loose finger clip design created inconsistent optical contact between the LEDs and the phototransistor, further decreasing signal quality. Eventually, by learning from these failed attempts and addressing the flaws, the final prototype operated efficiently.

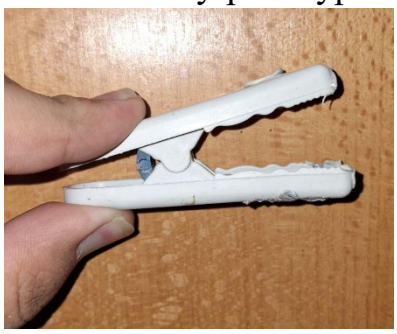


Figure 8: loose finger clip

Positive Results:

1. **Signal Quality:** After replacing the photodiode with a phototransistor and shielding the sensor from ambient light, the system produced a stable measurement.

2. **Heart-Rate Accuracy:** In three separate 30-second trials, the system achieved accurate heart rate readings of 90 BPM, 81 BPM, 79 BPM, 85 BPM , and 88 BPM meeting the design requirement compared to the commercial device, which had readings of 90 BPM,83 BPM, 81 BPM, 88 BPM, and 86 BPM.

3. **SpO_2 Measurement:** The improved setup allowed consistent oxygen saturation measurements of 99%, 98%, 99%,98%, and 97% across the three trials. On the other hand, the commercial device had readings of 99%, 97%, 98%, 99%, and 99% These results are summarized in figures 9, 10, and table 2.

Table 2: BPM and SPO_2 readings:

Trials	Ecopulse (BPM)	Commercial Device (BPM)	Ecopulse SpO_2 (%)	Commercial Device SpO_2 (%)
First trial	90	90	99%	99%
Second trial	81	83	98%	97%
Third trial	79	81	99%	98%
Fourth trial	85	88	98%	99%
Fifth trial	88	86	97%	99%

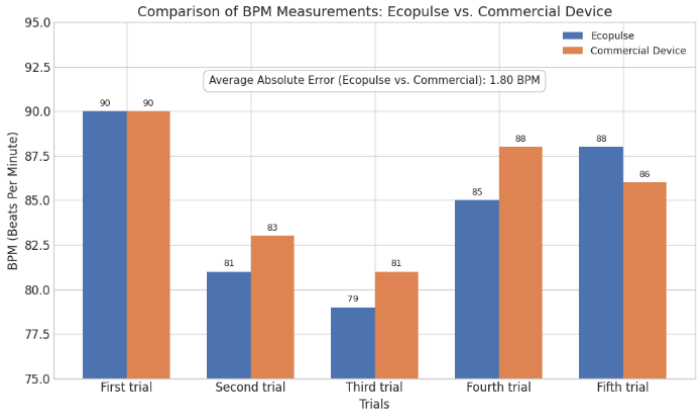


Figure 10: Comparison of BPM Measurements: Ecopulse vs. Commercial Device.

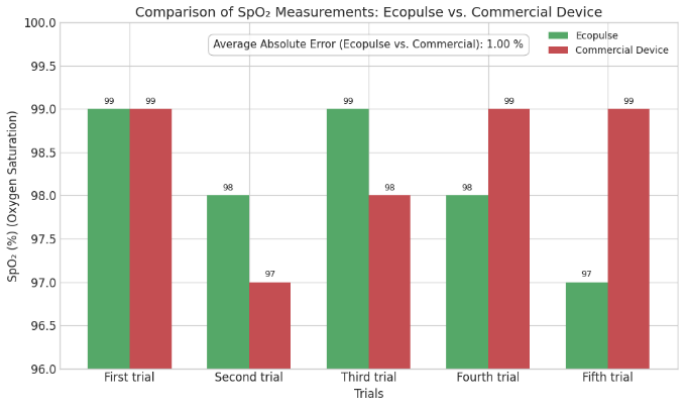


Figure 9: Comparison of SPO₂ Measurements: Ecopulse vs. Commercial Device.

Analysis

Table 3: Learning outcomes utilized during the project:

Learning Outcome	Description	Concepts	Usage
PH.2.03	Analyze current flow through conductors.	• Electric current • Voltage • Ohm's law	Used to calculate the necessary voltage drops and resistance for the LEDs and phototransistor using Ohm's Law.
PH.2.04	Analyze DC circuits with resistive elements.	• Connections of resistors • Kirchhoff's laws	Applied to design the circuit network ensuring the correct current flows through the phototransistor receiver.

MA.2.02	Model real situations with functions.	• Rational polynomial expressions • Modeling	Used to create a mathematical model linking light absorption ratios to the SpO_2 percentage readings.
PH.2.05	Analyze energy transfer in DC circuits.	• Power dissipation • Power ratings	Used to ensure recycled components operate within safe power limits to avoid overheating the finger clip.

Photoplethysmography

Photoplethysmography (PPG) is a simple optical method utilized in the detection of blood volume variations in the microvascular bed of tissue. The PPG signal works by illuminating the skin with pulses of light from LEDs, measuring the transmitted or reflected pulse of light detected by a photodetector. This generates tiny variations in the absorption of pulses of light based upon the pulsatile blood volumes in the cardiac cycle; this recording is referred to as the 'photoplethysmogram.'

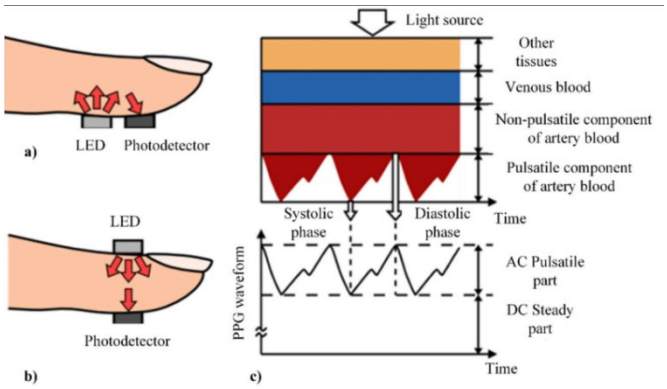


Figure 11: Photoplethysmogram mechanism diagram

The essence of PPG lies in the detection of the two major components: the pulsatile or alternating current (AC) synchronized with the heartbeat and the direct current (DC), referring to the overall absorption attributed to blood and tissue (a relatively stable baseline level).

The process of photoplethysmography (PPG) involves pulsing the skin tissue with red (around 660 nm) and infrared (around 880-940 nm) LEDs in periodic flickers—500 ms or less—and collecting the intensity of the transmitted light using a photodetector, as shown in Figure 11. The values of 660 nm and 880-940 nm are chosen based on the absorbance of each wavelength to the oxygen saturation in hemoglobin; oxygenated hemoglobin absorbs more infrared light and reflects red, which gives it a bright red color, whereas the deoxygenated hemoglobin absorbs more red light and reflects the wavelength closer to infrared, which gives it a deep dark red color. The measurements can be performed in either transmissive or reflective mode. The acquired PPG wave can be processed by converting it into the alternating or pulsatile part (AC) and the direct current (DC) part. The ratio of ratios (R) can be calculated from AC/DC ratios. The processed data reduces noise using digital filtering by Op-Amps.

The pulsatile-to-baseline ratio in the two wavelengths, referred to as R, is a fundamental value in the calculation of oxygen saturation measurements (SpO2). The ratio of ratios (R) is calculated using the difference in absorption between the

red and infrared pulses divided by the difference in infrared absorption, as shown in Equation 1. Because the relationship between SpO2 and R linearly inversely correlates with oxygenated hemoglobin absorption, the saturated oxygen percentage can be approximated by equations like Equation 2 and constrained between 80 and 100%. Although the PPG measurements are valid in many applications, their measurements may face small constraints due to different skin tones, resulting in errors between 2-4%.

The science behind the photoplethysmography concept was the backbone of building the pulse oximeter presented in this research and also has significant importance in the project's methodology: identifying the calculations, circuit connections, and even the materials used in implementing the idea.

$$R = \frac{AC_{RED}/DC_{RED}}{AC_{IR}/DC_{IR}}$$

Equation 1: The Ratio of Ratios

$$\mathrm{SpO_2} \approx 110 - 25\mathrm{R}$$

Equation 2: SpO2 Estimation Formula

Amplifying and noise reduction

The study of electrical components is essential to understanding the behavior of electric circuits under certain conditions and the limitations faced when building circuits. In this study, studying amplifiers, resistors, and capacitors is one of the fundamentals of constructing and building a photoplethysmograph (PPG).

Amplifiers are electrical components used in circuits to increase a signal's strength to a suitable level for processing, which are widely used in telephone wires. Without the use of amplifiers, signals coming from sensors, microphones, or photodiodes would be too small to process or detect, coming with micro- or millivolt outputs. In the PPG circuit, an LM358P op-amp was used to amplify the output signal from the phototransistor, as it fits perfectly with low voltage supplies (the 9V battery supply). Whereas other models like TL072 need higher outputs (± 10 Volts or more). So while the TL072 model offers more noise signal filtration and amplification, it is harder to include with the 9V output PPG circuit.

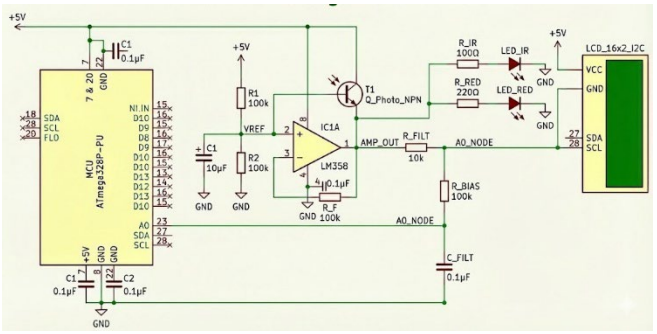


Figure 12: Schematic Circuit's Design

As shown in the schematic diagram illustrated in Figure 12, amplification isn't enough to maintain a clean, filtered psychological signal, but the use of additional capacitors and resistors throughout the circuit is essential. Resistors were used throughout the circuit,

starting from the 100-ohm resistors connected to the LEDs to avoid burning, the two 100-ohm resistors that were used to create the Vref node, and even the ones attached to the op-amp to reduce its signal. Additionally, capacitors, which are most essential for removing noise, were used as a backbone for the circuit in many connections, such as the A0 node, which is connected to the A0 pin in the AT mega; the Vref node, which is used to change the voltage interval for the PPG signal; and even between the Op-Amps pins to smooth out current.

Phototransistor vs. Photodiode

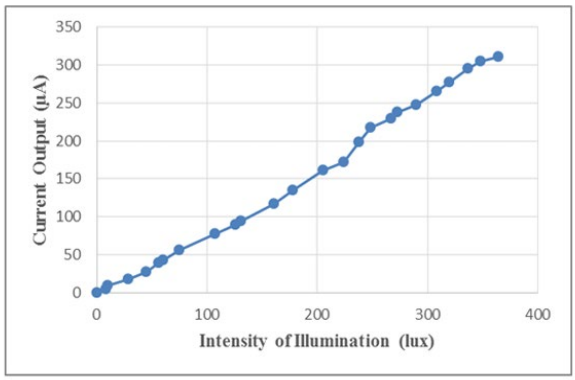


Figure 13: phototransistors current output to light intensity

Photosensors are designed to be sensitive to visible light or other electromagnetic radiation in a given spectral region of the electromagnetic spectrum that typically lies between ultraviolet and infrared radiation. Photodiodes operate primarily as a fast and highly responsive light detector that converts incident photons into electrical current through its simple p-n structure. This allows a very rapid response time, which makes photodiodes suitable for high-speed measurement. On the other hand, a phototransistor depends on internal current amplification, where the light generated is multiplied through transistor action. This amplification follows the relation

$$IC = \beta \cdot Iphoto$$

Equation 3: Phototransistor Current Amplification

where the small light-generated current is multiplied by the transistor gain β. Thus, very high sensitivity but a slower response time. It is basically a photodiode with a current transistor amplifier. So, it combines the characteristics of a transistor and a photodiode. For a heart rate and oxygen saturation monitoring system, a phototransistor was chosen because these biomedical sensing applications depend more on the signal strength under low light conditions than on extremely fast speeds. Figure 13 illustrates the phototransistor's high sensitivity, with the output current increasing from near zero to 300 μA as illumination rises to 400 lux. This pronounced current response under relatively low illumination supports its selection for heart-rate and SpO_2 measurements.

Accuracy Validation & Mechanical Optimization

The process of validating the accuracy and stability of the low-cost sensor was mainly focused on understanding how closely the readings follow the actual physiological values and how stable the signal behaves during measurement. The measurement bias was first considered since it shows the constant offset that appears when comparing the readings of the device to a calibrated commercial pulse

oximeter. The bias is calculated by taking the average difference between the measured values and the reference values, as shown in Equation 4, and it gives a clear idea about the consistency of the sensor in many trials.

$$Bias = \frac{1}{N}\sum_{i=1}^{N}\left(X_{device} - X_{reference}\right)$$

Equation 4: Measurement Bias

Using the deviation values recorded in Table 2, which were (0, –2, –2, –3, +2), the calculated bias reached -1.8 BPM. This small negative value shows that the readings tend to under-estimate the heart rate by about 1.8 BPM, which is considered very accurate in relation to low-cost optical systems, especially when taking into account the natural fluctuations of the pulsatile component. This bias also indicates that the waveform is being captured correctly with very low distortion from motion or tissue interference.

Another major part of the analysis was done through studying the signal stability by using the Signal-to-Noise Ratio (SNR). The SNR is very important because it reflects how much of the signal obtained is truly the pulsatile information and how much of it is unwanted noise, especially since the AC component of the PPG wave is usually very small compared to the DC level. The SNR is calculated using

Equation 5, which compares the power of the signal to the power of the noise and expresses it in decibels.

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

Equation 5: Signal-to-Noise Ratio

A transmissive finger-clip design was used because, based on recent studies, wrist-based reflective sensors normally suffer from a much lower SNR around 13.7 dB even when the subject is sitting still. This happens because reflective sensing depends only on the light bouncing from the top layer of the skin, which makes the readings very sensitive to any movements or changes in contact pressure. In contrast, transmissive sensing allows the light to travel through the tissue, which increases the amount of detected signal and reduces the contribution of noise from motion and external light.

Conclusions

This project addressed both the problem of resource constraints in Egyptian public-school laboratories and the challenge of how to dispose of or reuse e-waste. Using the principles of photoplethysmography, an operative model of a pulse oximeter that calculates heartbeats and blood-oxygen saturation rates was developed. Also, its development considered factors of sustainability through the

reuse of essential components such as infrared LEDs and photodiodes from disposed devices like remote controls or outdated Circuit boards. Indeed, it was assured that the developed model possessed all the requirements of the Capstone design challenge. It demonstrated accuracy in measurement with no more than a 5% margin of error, exactly **2.1**% relative to commercial standard pulse oximeters, which shows its effectiveness in the classroom setting. It further achieved its target in that it cost less than EGP 200, which is already far from the commercial prices ranging from EGP 500 to EGP 2,500.

This prototype provides a scalable solution for some of Egypt's grand challenges, like recycling garbage and waste for economic and environmental purposes. Compared to other commercial technologies that offer similar services but charge exorbitant prices unaffordable for most public schools, this innovative technology transforms hazardous electronic waste into a valuable tool that can enhance active learning. Furthermore, its portability allows for easy transport within most classrooms. In conclusion, the sustainable manufacturing of precision scientific instruments can be achieved through the development of such projects.

Recommendation

Empowering Education and Healthcare through Recycled Innovation

The eco-friendly heart rate and oxygen sensor addresses two critical challenges in Egypt: the lack of practical resources in STEM education and the scarcity of affordable medical equipment in rural healthcare. By repurposing electronic waste into functional scientific instruments, we transform an environmental hazard into a tool for social development.

In the educational sector, the device serves as a vital bridge between theoretical knowledge and practical application. Even advanced institutions, such as the STEM High School for Boys in 6th of October, face significant resource barriers. Currently, their biology labs rely primarily on static figures, diagrams, and microscopes, lacking the real scientific instruments required for dynamic experiments. With commercial sensors costing up to 2,500 EGP per unit, equipping a full class is often financially out of budget. Our recycled sensor provides a sustainable, low-cost alternative that allows students to visualize physiological concepts, like heart rate response to exercise, in real-time. This shifts the learning model from passive memorization to active scientific learning.



Figure 14: Real life location

Beyond the classroom, this device has profound applications in primary healthcare, particularly in underserved and rural communities. Many remote health units lack basic diagnostic tools due to high import costs. Our portable, battery-operated sensor offers an accessible solution for monitoring vital signs, including heart rate and oxygen saturation (SpO_2). It is ideal for use in mobile health caravans or for home monitoring of patients with chronic respiratory or cardiac conditions. By providing a reliable and inexpensive screening tool, we can facilitate early detection of health issues without the need for expensive hospital visits.

Ultimately, this project democratizes access to technology, ensuring that students in public schools and patients in rural villages have access to the same quality of education and healthcare monitoring as those in privileged urban centers.

Integration of IoT and Telemedicine Capabilities



Figure 15: A figure that shows how the IoT system works

The device works effectively as a standalone tool for real-time observation in the lab. However, a major limitation that was faced during testing is that it cannot store or share the data it collects. Once the user removes his finger, that valuable biological information is lost. To solve this and take the project to a professional level, it is strongly recommended to integrate a wireless communication module, such as the ESP8266 Wi-Fi chip, in future iterations. This

module is specifically recommended because it is cost-effective, widely available, and easily compatible with our current microcontroller setup.

Adding this technology would fundamentally transform the device from a simple school instrument into a sophisticated Internet of Things (IoT) node. Instead of just displaying numbers on a small LCD screen, the system could transmit patient data wirelessly to a dedicated mobile application or a cloud server database. This upgrade unlocks two massive opportunities. First, it enables telemedicine, allowing doctors to monitor a patient's vital signs remotely in real-time without requiring a physical hospital visit. Second, in the classroom, it will enable students to access long-term physiological data. They could download this history to computers to analyze heart rate trends over varying stress conditions, turning a basic biology lesson into a deep data analysis experience.

Cuffless Blood Pressure Estimation (PTT)

Although Pulse Transit Time (PTT) is recognized as a scientifically valid technique for blood pressure estimation, its implementation in this prototype was ruled out due to strict hardware and safety constraints. It

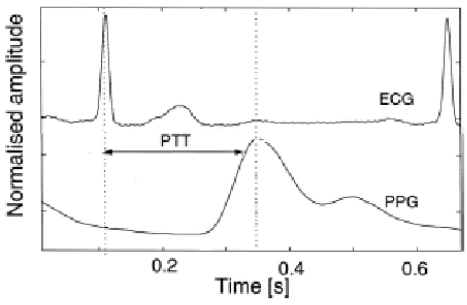


Figure 16: A graph that shows how to measure the PTT

was determined that the simultaneous, millisecond-precision sampling of ECG and PPG signals, which is required for accurate PTT, as shown in Figure 16, exceeds the processing limits of the single-core ATmega328P microcontroller used here.

Beyond processing power, a critical safety hazard is presented by the construction of an ECG front-end from scavenged e-waste. It was noted that such components lack the medical-grade electrical isolation necessary to protect the user from potential mains or USB faults. For future applications intended for real-life use, the adoption of a dual-core processor (like an ESP32 or STM32) is recommended to manage deterministic timing. Furthermore, it is suggested that a medically certified, optically isolated ECG module be integrated to ensure both the calculation accuracy and the electrical safety needed for human trials.

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